Charles Wolford

MSc, MBA, CGeog (GIS), FRGS

GIS Solutions Architect (Senior Associate), Digital Engineering, Ridge and Partners

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I build GIS capability that organisations run on, from technical strategy and platform architecture to the applications teams use every day. At Ridge, I'm responsible for GIS across the firm. At Amazon, I went from founding a new GIS programme as GIS Manager to leading the design and development of division-wide GIS infrastructure as GIS Solutions Architect. My career began in US Army geospatial intelligence (GEOINT) across conventional and special operations, where location data shaped decisions on the ground.

FOCUS AREAS GIS strategy and roadmaps · Enterprise and cloud GIS architecture · Digital estates · Asset condition platforms · Web and mobile applications · Data integration (BIM, CAD, drone imagery, IoT, APIs)

RELEVANT EXPERIENCE

SEP 2025 – PRESENT
Newcastle upon Tyne

Ridge and Partners LLP

GIS Solutions Architect (Senior Associate), Digital Engineering

- Oversees GIS solutions across Ridge from strategy to delivery as part of the Digital Engineering team; administers the firm's ArcGIS Online organisation and sets its GIS standards.
- Guides clients through critical GIS decisions and shapes digital estate roadmaps that move organisations from fragmented data to confident, evidence-led planning.
- Leads end-to-end delivery of interactive 2D and 3D estate experiences across healthcare, higher education and housing, uniting drone, survey, BIM and asset data in a single view: ArcGIS GeoBIM and floor plan pipelines for healthcare and science park estates (including resolution of Revit survey point georeferencing so models land correctly on the British National Grid), and the ArcGIS orthomosaic, 3D scene and defect mapping deliverable on drone condition surveys in collaboration with the Geospatial and Building Surveying teams.
- Builds and scales Ridge's GIS offer, from the platforms and services behind it to solutions tailored for each bid, including Experience Builder web applications with bespoke functionality, Survey123 and Field Maps mobile applications, and floor-aware mapping solutions for capacity management.

AUG 2023 – SEP 2025
London, England

Amazon

GIS Solutions Architect

- Technical lead for GIS within Amazon Corporate Security, guiding architecture, platform and tooling decisions.
- Led the design and development of the programme's GIS infrastructure, migrating it from ArcGIS Online to ArcGIS Enterprise hosted on AWS, and architected platforms that replaced multiple legacy solutions, delivering more than £2m in cost savings.
- Developed and deployed AI-powered features using AWS Bedrock and SageMaker, integrating large language models and machine learning into the platform.
- Trained corporate intelligence teams in Seattle, Dubai and London through hands-on GIS workshops, enabling them to map, analyse and act on location data independently.

APR 2022 – AUG 2023
London, England

Amazon

GIS Manager

- Rejoined Amazon to establish and lead a new GIS programme: built it from the ground up, defining strategy, priorities and roadmap, and secured buy-in across Corporate Security by turning operational needs into the programme's first GIS use cases.
- Launched Corporate Security's ArcGIS Online environment and managed its budget and licensing, setting the standards and governance that allowed it to scale.

FEB 2017 – JUN 2020
Prague, Czech Republic

Amazon

Senior Risk and Resilience Intelligence Analyst

- Led data-driven risk analysis and incident response using geospatial tools and statistical modelling, and built automated monitoring systems to identify and mitigate operational vulnerabilities across Amazon facilities.
- Developed real-time incident management dashboards and analytical frameworks that improved response times during critical events.

- Designed and delivered training programmes, technical documentation and hands-on workshops across Amazon sites globally to build spatial analysis capability and drive adoption of data visualisation tools.

JUN 2008 – JUL 2015
Global operations

United States Army

Operations Manager and Senior Geospatial Intelligence Analyst

- Directed daily operations, training and professional development for a team of up to 60 personnel while serving as Senior GEOINT Analyst supporting United States Special Operations Command (SOCOM) and Special Operations Command Africa (SOCAF).
- Led geospatial intelligence production for special operations forces, NATO and national-level agencies, overseeing more than 1,400 imagery products and over 1,100 hours of full motion video analysis.
- Brought GIS into frontline intelligence, maintaining daily route layers of IEDs, checkpoints and points of interest, and fusing imagery with signals and human intelligence into a single operational picture.
- Held a Top Secret/SCI clearance with polygraph and completed specialised training including Distributed Ground System, Special Operations Forces (DGS-SOF), AFSOC Full Motion Video analysis certification and Joint Special Operations Task Force certification.

EDUCATION

University of Aberdeen	**MSc Geographical Information Systems.** Spatial mathematics, geodetic science, computer science, database systems, and computational analysis of geographic data.
Quantic School of Business and Technology	**Executive MBA.** Specialisations in strategic operations, organisational leadership and financial management.
University of New York in Prague	**BA Communication and Media.** S.A.L.U.T.E. Veterans National Honor Society graduate; specialisations in management and marketing.

PROFESSIONAL ACCREDITATION AND CERTIFICATIONS

Royal Geographical Society	Chartered Geographer (GIS), CGeog (GIS), accredited 2022
Royal Geographical Society	Fellow, FRGS
Amazon Web Services	AWS Certified AI Practitioner; AWS Certified Cloud Practitioner
Google Cloud	Google Cloud Digital Leader
US Army Intelligence and Security Command	Diploma in Geospatial Intelligence Analysis
Other	Governors for Schools, School Governor; ASIS International, Member

TECHNICAL CAPABILITY

Esri platform	ArcGIS Enterprise (Portal, Server, Data Store) deployment and administration on AWS, ArcGIS Online administration, ArcGIS Pro, ArcGIS GeoBIM, Experience Builder, Survey123, ArcGIS Solutions, ArcGIS Maps for Power BI, ArcGIS Indoors, ArcGIS Field Maps, ArcGIS Dashboards, ArcGIS Workflow Manager, ArcGIS API for Python, ArcGIS Velocity
Data integration and geodesy	Spatial data modelling and schema governance, metadata, coordinate reference systems and transformations (British National Grid, OSGB36, ETRS89, OSTN15), CAD and BIM to GIS, point cloud and drone imagery products, PostGIS
Development	Python, Arcade, JavaScript, R, SQL, NoSQL, REST APIs, Docker, Kubernetes, Terraform, Git
Cloud and AI	AWS (S3, EC2, CloudFront, Bedrock, SageMaker, Glue, Lambda, Redshift, RDS, IAM, VPC and more), Google Cloud Platform; machine learning, natural language processing, predictive modelling
Leadership	Technical strategy and programme establishment, client advisory, governance and standards, stakeholder engagement, programme management, budget and licensing management, risk management, quality assurance and continuous improvement, capability building and documentation, team leadership

RIGHT TO WORK AND SECURITY

Right to work in the United Kingdom, European Union and United States. UK resident since October 2020.